Informatica® Cloud Data Integration

Open Table Connector

Informatica Cloud Data Integration Open Table Connector
July 2025

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Preface

Use *Open Table Connector* to learn how to read from or write to Apache Iceberg or Delta Lake tables by using Cloud Data Integration. Learn to create an Open Table connection, develop and run mappings, and mapping tasks in Cloud Data Integration.

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CHAPTER 1

Introduction to Open Table Connector

You can use Open Table Connector to connect to a catalog from Data Integration to read from and write data to Open Table formats.

You can create an Open Table connection and use it in mappings to securely read from or write data to Apache Iceberg tables managed by the AWS Glue catalog, Polaris REST catalog, or Nessie catalog.

When you run mappings in advanced mode, you can use an Open Table connection to securely read from or write data to Delta Lake Open Table formats managed by the AWS Glue catalog or Apache Iceberg Open Table formats managed by the AWS Glue catalog , Hive Metastore, Polaris REST catalog, or Nessie catalog.

Open Table Connector assets

Create assets in Data Integration to integrate data using Open Table Connector.

When you use Open Table Connector, you can include the following Data Integration assets:

- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Connections for Open Table

Create an Open Table connection to securely read from or write data to Open Table formats available in a catalog.

You can use an Open Table connection to specify sources, targets, and lookups in mappings and mapping tasks.

Prerequisites

Before you create an Open Table connection, complete the prerequisites.

Using AWS Glue Catalog and Amazon S3 Storage to interact with Apache Iceberg or Delta Lake tables

If you use an AWS Glue Catalog and Amazon S3 Storage to interact with Apache Iceberg or Delta Lake tables, you need to have access to the following AWS services that manage the tables on AWS:

- AWS Glue Catalog. AWS Glue Catalog manages the metadata associated with the Apache Iceberg or Delta Lake tables.
- Amazon S3 Storage. Amazon S3 stores the metadata files and data files of the Apache Iceberg or Delta Lake tables.
- Amazon Athena. Amazon Athena uses the AWS Glue Data Catalog to store metadata such as table and column names for your data stored in Amazon S3. Open Table Connector uses the Amazon Athena JDBC driver to connect to the AWS Glue Catalog to access Apache Iceberg or Delta Lake tables metadata.

You need to create separate policies to access these services.

Using Hive Metastore catalog and Microsoft Azure Data Lake Storage Gen2 to interact with Apache Iceberg tables

If you use a Hive Metastore catalog and Microsoft Azure Data Lake Storage Gen2 to interact with Apache Iceberg tables, you need to have access to the following services that manage the tables on Microsoft Azure Data Lake Storage Gen2:

- Hive Metastore catalog. Hive Metastore catalog manages the metadata associated with the Apache Iceberg tables. The metastore must use Hive version 4.0.
- Microsoft Azure Data Lake Storage Gen2. Microsoft Azure Data Lake Storage Gen2 stores the metadata files and data files of the Apache Iceberg tables.
- Hive JDBC driver. Hive JDBC driver connects to the Hive server to access the metadata of Apache Iceberg tables.

Using Hive Metastore catalog and Amazon S3 storage to interact with Apache Iceberg tables

If you use a Hive Metastore catalog and Amazon S3 storage to interact with Apache Iceberg tables, you need to have access to the following services that manage the tables on Amazon S3 storage:

- Hive Metastore catalog. Hive Metastore catalog manages the metadata associated with the Apache Iceberg tables. The metastore must use Hive version 4.0.
- Amazon S3 storage. Amazon S3 stores the metadata files and data files of the Apache Iceberg tables.
- Hive JDBC driver. Hive JDBC driver connects to the Hive server to access the metadata of Apache Iceberg tables.

Using REST catalog and Amazon S3 to interact with Apache Iceberg tables

If you use a REST catalog such as Polaris catalog and Amazon S3 storage to interact with Apache Iceberg tables, you need to have access to the following services that manage the tables on Amazon S3 storage:

- REST catalog. REST catalog manages the metadata associated with the Apache Iceberg tables.
- Amazon S3 storage. Amazon S3 stores the metadata files and data files of the Apache Iceberg tables.

Note: To list all catalogs in a Polaris REST catalog, ensure that the Principal user has the `service_admin` privileges.

Using REST catalog and Microsoft Azure Data Lake Storage Gen2 to interact with Apache Iceberg tables

If you use a REST catalog such as Polaris catalog and Microsoft Azure Data Lake Storage Gen2 to interact with Apache Iceberg tables, you need to have access to the following services that manage the tables on Microsoft Azure Data Lake Storage Gen2:

- REST catalog. REST catalog manages the metadata associated with the Apache Iceberg tables.
- Microsoft Azure Data Lake Storage Gen2. Microsoft Azure Data Lake Storage Gen2 stores the metadata files and data files of the Apache Iceberg tables.

Note: To list all catalogs in a Polaris REST catalog, ensure that the Principal user has the `service_admin` privileges.

Using Nessie catalog and MinIO to interact with Apache Iceberg tables

If you use a Nessie catalog and MinIO storage to interact with Apache Iceberg tables, you need to have access to the following services that manage the tables on MinIO storage:

- Nessie catalog. Nessie catalog manages the metadata associated with the Apache Iceberg tables.
- MinIO storage. MinIO stores the metadata files and data files of the Apache Iceberg tables.

Create minimal IAM policies

You need to create IAM policies with the minimum required permissions to interact with Apache Iceberg or Delta Lake tables managed by AWS Glue Catalog. For more information on configuring these policies, refer to the AWS documentation.

Minimum policy for Amazon Athena

The following sample policy shows the minimal Amazon IAM policy to access Amazon Athena:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
```

```
            "Action": [
                "athena:CreatePreparedStatement",
                "athena:GetPreparedStatement",
                "athena:GetWorkGroup",
                "athena:GetTableMetadata",
                "athena:StartQueryExecution",
                "athena:GetQueryResultsStream",
                "athena:ListDatabases",
                "athena:GetQueryExecution",
                "athena:GetQueryResults",
                "athena:GetDatabase",
                "athena:ListTableMetadata",
                "athena:GetDataCatalog",
                "athena:DeletePreparedStatement"
            ],
            "Resource": [
                "arn:aws:athena:*:*:workgroup/*",
                "arn:aws:athena:*:*:datacatalog/*"
            ]
        },
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "athena:ListDataCatalogs",
                "athena:GetQueryExecution",
                "athena:ListWorkGroups",
                "athena:GetPreparedStatement"
            ],
            "Resource": "*"
        }
    ]
}
```

Minimum policy for AWS Glue

The following sample policy shows the minimal Amazon IAM policy to access AWS Glue Catalog:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "glue:*"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Minimum policy for AWS S3

The following sample policy shows the minimal Amazon IAM policy to read from or write data to an Amazon S3 bucket:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:ListBucket",
                "s3:DeleteObject"
            ],
            "Resource": "*"
```

```
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:ListAllMyBuckets",
                "s3:GetBucketAcl"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Install the Hive JDBC driver for Hive metastore

Before you use Open Table Connector, you need to copy the Hive JDBC driver on the Linux machine where you installed the Secure Agent. You need to use the Hive JDBC driver for the Hive Metastore catalog.

1. Download the latest Hive JDBC driver from the website.
2. Navigate to the following directory on the Secure Agent machine: `<Secure Agent installation directory>/ext/connectors/thirdparty/`
3. Create the following folder: `informatica.opentableformat/common`
4. Add the JDBC driver to the folder.
5. Restart the Secure Agent.

Configure EC2 role to assume role

You can configure an EC2 role to assume an IAM role and generate temporary security credentials to connect to Amazon S3 from the same or different AWS accounts.

When you configure EC2 role to assume role, ensure that you have the **sts:AssumeRole** permission and a trust relationship established within the AWS accounts to use the temporary security credentials. The trust relationship is defined in the trust policy of the IAM role when you create the role. The IAM role adds the EC2 role as a trusted entity allowing the EC2 role to use the temporary security credentials and access the AWS accounts.

When the trusted EC2 role requests for the temporary security credentials, the AWS Security Token Service (AWS STS) dynamically generates the temporary security credentials that are valid for a specified period and provides the credentials to the trusted EC2 role.

Before you use the EC2 Role to Assume Role authentication, consider the following prerequisites:

- Install the Secure Agent on the AWS EC2 instance.
- The EC2 role attached to the AWS EC2 instance must have permissions to assume another IAM role. The following is a sample permission policy of EC2 role that is attached to the AWS EC2 instance:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "sts:AssumeRole",
            "Resource": "arn:aws:iam::001234567890:role/open_table_rolearn"
        }
    ]
}
```

Resource value must include the ARN of IAM role that the EC2 role needs to assume.

- The IAM role that the EC2 role needs to assume must have a permission policy and a trust policy attached to access AWS Glue Catalog, Amazon Athena, and Amazon S3.
  You can also specify the external ID of your AWS account for a more secure access. The external ID must be a string.

  The following sample shows the assumed IAM role's trust policy with the external ID:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::001234567890:root" //anyone in this account
001234567890 can assume this role, this can also be limited to one role.
            },
            "Action": "sts:AssumeRole",
            "Condition": {
                "StringEquals": {
                    "sts:ExternalId": "aws_externalid"
                }
            }
        }
    ]
}
```

For more information about the minimum permission policies, see "Create minimal IAM policies" on page 9.

Configure cross-account access

You can configure cross-account access when the Secure Agent running on an EC2 instance is in one AWS account, while Apache Iceberg tables, AWS Glue Catalogs, and Athena workgroups reside in a different AWS account.
Consider the following example to set up cross-account IAM roles and policies to allow an EC2 agent running in Account B to access AWS Glue Catalog, Athena workgroup, and Apache Iceberg tables located in Account A:

- Account A hosts the IAM Role to be assumed, AWS Glue Catalog, Athena workgroup, and Iceberg tables.
- Account B hosts the EC2 instance running the Secure Agent that needs to assume the role in Account A.

Perform the following steps to configure cross-account access for an EC2 instance:

1. Create IAM policies in Account A for AWS Glue Catalog, Amazon Athena, and Amazon S3.
2. Create an IAM role in Account A with trust policy allowing Account B to assume the role using an External ID.
3. Create an IAM policy in Account B allowing EC2 instance to assume the role in Account A.
4. Create and assign the IAM Role to the EC2 instance in Account B.

Create IAM policies in Account A

You need to create IAM policies with the minimum required permissions to interact with AWS Glue Catalog, Amazon Athena, and Amazon S3.

1. Log in to the IAM console.
2. Click **Policies** > **Create policy**.
3. On the **JSON** tab, add the following policies:

Minimum policy for Amazon Athena

The following sample policy shows the minimal Amazon IAM policy to access Amazon Athena:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "athena:CreatePreparedStatement",
                "athena:GetPreparedStatement",
                "athena:GetWorkGroup",
                "athena:GetTableMetadata",
                "athena:StartQueryExecution",
                "athena:GetQueryResultsStream",
                "athena:ListDatabases",
                "athena:GetQueryExecution",
                "athena:GetQueryResults",
                "athena:GetDatabase",
                "athena:ListTableMetadata",
                "athena:GetDataCatalog",
                "athena:DeletePreparedStatement",
                "athena:ListWorkGroups",
                "athena:ListDataCatalogs"
            ],
            "Resource": [
                "arn:aws:athena:*:*:workgroup/*",
                "arn:aws:athena:*:*:datacatalog/*"
            ]
        },
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "athena:ListDataCatalogs",
                "athena:GetQueryExecution",
                "athena:ListWorkGroups",
                "athena:GetPreparedStatement"
            ],
            "Resource": "*"
        }
    ]
}
```

Minimum policy for AWS Glue

The following sample policy shows the minimal Amazon IAM policy to access AWS Glue Catalog:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "glue:*"
            ],
            "Resource": "*"
        }
    ]
}
```

Minimum policy for Amazon S3

The following sample policy shows the minimal Amazon IAM policy to read from or write data to an Amazon S3 bucket:

```
{
    "Version": "2012-10-17",
```

```
        "Statement": [
            {
                "Sid": "VisualEditor0",
                "Effect": "Allow",
                "Action": [
                    "s3:PutObject",
                    "s3:GetObject",
                    "s3:ListBucket",
                    "s3:DeleteObject"
                ],
                "Resource": "*"
            },
            {
                "Effect": "Allow",
                "Action": [
                    "s3:ListBucket",
                    "s3:GetBucketLocation",
                    "s3:ListAllMyBuckets",
                    "s3:GetBucketAcl"
                ],
                "Resource": [
                    "*"
                ]
            }

        ]
    }
```

Create IAM Role in Account A

Create an IAM role in Account A with trust policy allowing Account B to assume the role using an External ID.

1. In the IAM console, click **IAM** > **Roles** > **Create Role**.
2. On the **Step 1: Select trusted entities** tab, under the **Trusted entity type** section, select **AWS account**.
3. In the **An AWS account** section, select **Another AWS account**, and enter the Account ID for Account B that hosts the EC2 instance where the Secure Agent is installed.
4. Under **Options**, select **Require External ID** and enter a secure external ID.
5. Click **Next**.
6. On the **Step 2 Add permissions** tab, select and attach the created permission policies to access AWS Glue Catalog, Amazon Athena, and Amazon S3.
7. Click **Next**.
8. On the **Step 3 Name, review, and create** tab, enter a **Name** for the role.
9. Click **Create role**.
   After you create a role, copy the IAM role ARN. You must specify the IAM role ARN in the Open Table connection to use cross-account access for EC2 Role to Assume Role authentication.
10. On the **Trust relationships** tab, you can view trust relationship policy for the IAM role.
    The following example shows the trust relationship policy for the IAM role:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": [
                    "arn:aws:iam::123456789101:root"
                ]
            },
            "Action": "sts:AssumeRole",
            "Condition": {
```

```
                "StringEquals": {
                    "sts:ExternalId": "opentable"
                }
            }
        }
    ]
}
```

Create an IAM policy in Account B

Create an IAM policy in Account B allowing the EC2 instance to assume the IAM role in Account A.

1. In the IAM console, click **Policies** > **Create policy**.
2. On the **JSON** tab, add the following policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": "sts:AssumeRole",
            "Resource": "arn:aws:iam::AccountAID:role/assumerolename"
        }
    ]
}
```

   The value of the **Resource** field in the policy is the ARN of the IAM Role created in Account A. Replace `arn:aws:iam::AccountAID:role/assumerolename` with the actual IAM role ARN.

3. Click **Next**.
4. Name the policy, and then click **Create Policy**.

Create and assign the IAM Role to the EC2 instance in Account B

To create an IAM role for the EC2 instance in Account B, perform the following steps:

1. In the IAM console, click **IAM** > **Roles** > **Create Role**.
2. On the **Step 1: Select trusted entities** tab, under the **Trusted entity type** section, select **AWS service**.
3. Under **Use case**, select **EC2**.
4. Click **Next**.
5. On the **Step 2 Add permissions** tab, select and attach the created policy in Account B.
6. Click **Next**.
7. On the **Step 3 Name, review, and create** tab, enter the **Name** and **Description** for the role.
8. Click **Create role**.

To assign the IAM Role to the EC2 instance, perform the following steps:

1. Log in to the EC2 Console.
2. Select your instance, and then click **Actions** > **Security** > **Modify IAM role**.
3. Select the role you created, and then click **Update IAM role**.

Optionally, to whitelist an IP address, perform the following steps:

1. On the **Security** tab, click on the ID of the attached **Security Group**.
2. Click **Edit inbound rules**.
3. Click **Add rule**, select the **Protocol** and **Port range**.
4. Under **Source**, select **Custom** and enter the specific IP address you want to whitelist.

5. Click **Save rules**.

Connect to Open Table

Let's configure the Open Table connection properties to connect to AWS Glue Catalog, Hive Metastore, Polaris REST catalog, or Nessie catalog.

Before you begin

Before you get started, you will need to create the minimal IAM policies to interact with Apache Iceberg or Delta Lake tables managed by AWS Glue Catalog and install the Hive JDBC driver for Hive metastore. You also need to configure the authentication-specific prerequisites to connect to Amazon S3 or Microsoft Azure Data Lake Storage Gen2 storage.

Permanent IAM Credentials authentication for Amazon S3 requires the access key and secret key values of the IAM user. EC2 Role to Assume Role authentication for Amazon S3 requires the ARN of the IAM role that the EC2 role assumes to generate temporary security credentials.

To configure Service Principal authentication for Microsoft Azure Data Lake Storage Gen2, you need the Azure account name, client secret, client ID, and tenant ID for your application registered in the Azure Active Directory.

Check out "Prerequisites" on page 8 to learn more about how to configure policies and role to access Apache Iceberg or Delta Lake tables.

Open Table formats with associated catalog and storage types

You can choose the Open Table format that you want to use and its associated catalog type and storage type to interact with data.

The following table summarizes the Open Table formats that you can use, their catalog types, storage types, and the authentication options available for each storage type:

Open Table format	Catalog type	Catalog authentication type	Storage type	Storage authentication type
Apache Iceberg	AWS Glue Catalog*	None	Amazon S3*	- AWS Permanent IAM Credentials authentication - EC2 Role to Assume Role
	Hive Metastore	None	Amazon S3	AWS Permanent IAM Credentials authentication
	Hive Metastore	None	Microsoft Azure Data Lake Storage Gen2	Azure Service Principal authentication
	REST Catalog*	OAuth 2.0 Credentials	Amazon S3*	- AWS Permanent IAM Credentials authentication - Credential Vending

Open Table format	Catalog type	Catalog authentication type	Storage type	Storage authentication type
	REST Catalog	OAuth 2.0 Credentials	Microsoft Azure Data Lake Storage Gen2	- Azure Service Principal authentication - Credential Vending
	Nessie Catalog*	OAuth 2.0 Credentials	MinIO*	MinIO Permanent IAM Credentials authentication
Delta Lake	AWS Glue Catalog	None	Amazon S3	AWS Permanent IAM Credentials authentication
*Open Table formats with catalog and storage types that apply to both mappings and mappings in advanced mode. The other catalog and storage types apply only to mappings in advanced mode.				

Connection details

The following table describes the Open Table connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent, Hosted Agent, or serverless runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Open Table Format	The Open Table format that you want to use to read from or write data to a catalog. Select Apache Iceberg or Delta Lake from the list.

Catalog types

You can select AWS Glue Catalog, Hive Metastore, REST Catalog, or Nessie catalog as the catalog type to manage the metadata of the Open Table format that you selected.

Select the catalog type that your Open Table format uses and then configure the catalog specific parameters.

AWS Glue Catalog

If the Apache Iceberg or Delta Lake Open Table format uses AWS Glue Catalog as the catalog type, configure the properties specific to AWS Glue Catalog.

The following table describes the property to configure AWS Glue Catalog:

Property	Description
Athena JDBC URL	Enter the JDBC URL in the following format: `jdbc:athena://Region=<AWS_Region>;OutputLocation=<S3_Location>` For example, `jdbc:athena://Region=us-west1;OutputLocation=s3://working/dir`.
Catalog Authentication Type	The authentication method to connect to the catalog. Select one of the following options: - None. Connects to AWS Glue Catalog or Hive Metastore without any authentication credentials. - OAuth 2.0 Client Credentials. Connects to a REST catalog or Nessie catalog using a Client ID and Client Secret to obtain an access token from the authorization server.

Hive Metastore

If the Apache Iceberg Open Table format uses Hive Metastore as the catalog type, configure the properties specific to Hive Metastore.

The following table describes the properties to configure Hive Metastore:

Property	Description
Hive Metastore URI	The Hive thrift server URL to connect to Hive Metastore.
Hive JDBC URL	The JDBC URL to connect to Hive4 server.
Hive User Name	The user name of your Hive account to connect to Hive Metastore.
Hive Password	The password of your Hive account to connect to Hive Metastore.
Catalog Authentication Type	The authentication method to connect to the catalog. Select one of the following options: - None. Connects to AWS Glue Catalog or Hive Metastore without any authentication credentials. - OAuth 2.0 Client Credentials. Connects to a REST catalog or Nessie catalog using a Client ID and Client Secret to obtain an access token from the authorization server.

REST Catalog

If the Apache Iceberg Open Table format uses REST catalog as the catalog type, configure the properties specific to REST Catalog.

The following table describes the properties to configure REST catalog:

Property	Description
REST Catalog Type	The type of REST catalog that you want to connect to. Select Polaris Catalog.
Catalog Endpoint URL	The endpoint URL of the REST catalog. For example, `http://35.98.240.22:8181`.
Catalog Authentication Type	The authentication method to connect to the catalog. Select one of the following options: - None. Connects to an AWS Glue Catalog or a Hive Metastore without any authentication credentials. - OAuth 2.0 Client Credentials. Connects to a REST catalog or Nessie catalog using a client ID and client secret to obtain an access token from the OAuth 2.0 authorization server.
Access Token URL	The URL provided by the OAuth 2.0 authorization server to obtain an access token. For example, `http://35.98.240.22:8181/api/catalog/v1/oauth/tokens`.
Client ID	The client ID of OAuth 2.0 authorization server that is registered with the catalog.
Client Secret	The client secret of OAuth 2.0 authorization server that is registered with the catalog.
Scope	The scope parameters that define the permissions an access token grants to the catalog. For example, `PRINCIPAL_ROLE:ALL`.

Nessie Catalog

If the Apache Iceberg Open Table format uses Nessie catalog as the catalog type, configure the properties specific to Nessie Catalog.

The following table describes the properties to configure Nessie catalog:

Property	Description
Nessie URI	The base URI of the Nessie REST API endpoint. For example, `http://3.8.140.58:18120/api/v2`.
Catalog Authentication Type	The authentication method to connect to the catalog. Select one of the following options: - None. Connects to an AWS Glue Catalog or a Hive Metastore without any authentication credentials. - OAuth 2.0 Client Credentials. Connects to a REST catalog or Nessie catalog using a client ID and client secret to obtain an access token from the OAuth 2.0 authorization server.
Access Token URL	The URL provided by the OAuth 2.0 authorization server to obtain an access token. For example, `https://in.okta.com/oauth2/default/v1/token`.
Client ID	The client ID of OAuth 2.0 authorization server that is registered with the catalog.

Property	Description
Client Secret	The client secret of OAuth 2.0 authorization server that is registered with the catalog.
Scope	The scope parameters that define the permissions an access token grants to the catalog. For example, `nessieopentablepyn.access`.

Storage types

You can choose Amazon S3 , Microsoft Azure Data Lake Storage Gen2, or MinIO as the storage type to store the Open Table format tables.

Select the storage type and configure the storage specific authentication parameters.

Amazon S3

If you use AWS Glue Catalog, Hive Metastore, or REST Catalog as the catalog type, configure the properties specific to Amazon S3 storage.

AWS Permanent IAM Credentials authentication

You can use AWS Permanent IAM Credentials authentication for Amazon S3 storage when you connect to an AWS Glue Catalog, Hive Metastore, or REST Catalog.

The following table describes the properties to configure AWS Permanent IAM Credentials authentication:

Property	Description
Access Key	The key to access the AWS Glue Catalog.
Secret Key	The secret key to access the AWS Glue Catalog. The secret key is associated with the access key and uniquely identifies the account.

AWS EC2 Role to Assume Role authentication

You can use AWS EC2 Role to Assume Role authentication for Amazon S3 storage only when you read Apache Iceberg tables from AWS Glue Catalog.

The following table describes the properties to configure AWS EC2 Role to Assume Role authentication:

Property	Description
IAM Role ARN	The ARN of the IAM role assumed by the EC2 role to generate the temporary session credentials.
External ID	A unique, user-defined string value that the IAM role requires the EC2 role to provide when calling the sts:AssumeRole API.

Credential Vending

You can use credential vending for Amazon S3 storage when you connect to a REST Catalog.

Credential vending determines whether the storage for the REST catalog requires authentication. When you select credential vending, the REST catalog automatically generates temporary session credentials and handles storage access. You do not need to provide the storage credentials separately.

Microsoft Azure Data Lake Storage Gen2

If you use Hive Metastore or REST Catalog as the catalog type, configure the properties specific to Microsoft Azure Data Lake Storage Gen2.

Azure Service Principal authentication

The following table describes the properties to configure Service Principal authentication:

Property	Description
Azure Account Name	The name of the Microsoft Azure Data Lake Storage Gen2 account to stage the files.
Azure Client ID	The client ID of your application. Enter the application ID or client ID for your application registered in the Azure Active Directory.
Azure Client Secret	The client secret for your application.
Azure Tenant ID	The directory ID or tenant ID for your application.

Credential Vending

You can use credential vending for Microsoft Azure Data Lake Storage Gen2 when you connect to a REST Catalog.

Credential vending determines whether the storage for the REST catalog requires authentication. When you select credential vending, the REST catalog automatically generates temporary session credentials and handles storage access. You do not need to provide the storage credentials separately.

MinIO

If you use Nessie catalog as the catalog type, configure the properties specific to MinIO.

Select MinIO Permanent IAM Credentials authentication as the authentication type to access Open Table formats in MinIO.

MinIO Permanent IAM Credentials authentication

The following table describes the properties to configure MinIO Permanent IAM Credentials authentication:

Property	Description
Endpoint URL	The HTTP or HTTPS URL of the MinIO storage endpoint. For example, `http://3.8.140.58:9000`.
Access Key	Access key ID that uniquely identifies the IAM user credentials to access MinIO.
Secret Key	Secret access key that authenticates the access key ID to securely access MinIO.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Warehouse Location	The path to the warehouse in MinIO where the Nessie catalog stores the Iceberg tables. For example, `s3://warehouse/`.
Additional Properties	Additional properties to configure the MinIO storage. The default properties are `client.region=us-east-1` and `s3.path-style-access=true`. Do not edit or delete the `s3.path-style-access=true` property. You can edit the `client.region` property and set the value to the MinIO region for your catalog. Enter new properties as key=value pairs separated by semicolons.

CHAPTER 3

Mappings and mapping tasks with Open Table Connector

Use the Data Integration Mapping Designer to define and configure a mapping for Open Table sources and targets.

In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

When you run a mapping in advanced mode, Data Integration analyzes the data logic defined in the mapping to automatically assign data logic to run on an advanced cluster.

You can use Monitor to monitor the job.

When you run a mapping task and it completes, you can view the count of processed target rows on the job summary page. You can also view the count of processed and failed rows for each task on the Individual Task Results page. The runtime statistics show information of the read and insert operations.

Open Table sources in mappings

When you configure a mapping to use an Open Table source, you can configure the source properties.

Specify the name and description of the Open Table source. Configure the source and advanced source properties for the Open Table object.

The following table describes the Open Table source properties that you can configure in a Source transformation:

Property	Description
Connection	Name of the source connection. You can select an existing connection, create a new connection, or define parameter values for the source connection property. If you want to overwrite the source connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Object Type	Type of the Open Table source object. You can choose from the following source types: - Single Object. Select to specify a single Open Table object. - Parameter. Select to specify a parameter name. You can configure the source object in a mapping task associated with a mapping that uses this Source transformation. - Query*. Select to perform a complicated join of multiple tables or to reduce the number of fields that enter the data flow in a very large source.
Parameter	A parameter file where you define values that you want to update without the need to edit the task. Select an existing parameter for the source object or click **New Parameter** to define a new parameter for the source object. The **Parameter** property appears only if you select parameter as the source type. If you want to overwrite the parameter at run time, select the **Allow parameter to be overridden at run time** option. When the task runs, the Secure Agent uses the parameters from the file that you specify in the advanced session properties.
Object	Source object for the mapping.
Query*	This property appears only if you select Query as the object type. Click **Define Query** and enter a valid custom query. You can parameterize a custom query object at run time in a mapping.
*Doesn't apply to mappings in advanced mode.	

The following table describes the Open Table query options that you can configure in a Source transformation:

Property	Description
Filter	Filter value in a read operation. Click **Configure** to add conditions to filter records and reduce the number of rows that the Secure Agent reads from the source. You can specify the following filter conditions: - **Not Parameterized**. Use a basic filter to specify the object, field, operator, and value to select specific records. - **Completely Parameterized**. Use a parameter to represent the field mapping. - **Advanced**. Use an advanced filter to define a more complex filter condition that uses the Open Table query format.
Sort	Not applicable

The following table describes the Open Table source advanced properties that you can configure in a Source transformation:

Property	Description

Iceberg Advanced Properties	The properties that you want to configure for the Iceberg tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	In a mapping, when you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<bucket-region-name>`
	In a mapping in advanced mode, when you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket ARN property in the following format:
	`s3.access-points.<bucket-name>=<S3-bucket-ARN>`
	For more information about Amazon S3 bucket ARN, see the Create the ARN for your AWS S3 bucket Knowledge Base article.
	When you use Hive metastore with Amazon S3, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<Amazon-S3-bucket-region-name>`
	When you use REST Catalog with Amazon S3, you must specify the Amazon S3 bucket region property in the following format:
	`restcatalog.iceberg.s3.client.region=<Amazon-S3-bucket-region-name>`

Delta Spark Properties*	The properties that you want to configure for the Delta Lake tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<bucket-region-name>`

Pre-SQL	Pre-SQL queries to run before reading data from Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the pre-SQL query fails, the mapping also fails.
Post-SQL	Post-SQL queries to run after reading data from Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the post-SQL query fails, the mapping also fails. If the mapping fails, the post-SQL query is not executed.

Property	Description

Time Travel Query	Time travel query to fetch historical data of a table based on a timestamp or snapshot ID you specify.
	Enter a value for snapshot ID or timestamp in the following format:
	`SnapshotVersion=<snapshot value, latest, or latest-N>` OR
	`TimestampValue=<YYYY-MM-DD HH:mm:ss.SSS UTC, latest, or latest-N>`
	For example, you can specify latest as your timestamp value to fetch the recent generated snapshot:
	`SnapshotVersion=latest`
	You can partially parameterize the time travel query using an input parameter and resolve the parameter in the parameter file.
	For more information about configuring Time Travel Query, see "Time Travel Query" on page 26.

Tracing Level	Sets the amount of detail that appears in the log file. You can choose terse, normal, verbose initialization, or verbose data. Default is normal.
*Applies only to mappings in advanced mode.	

Time Travel Query

You can use the time travel query to roll back historical data of a table.

When you add or delete any data in Apache Iceberg table items, it automatically generates a snapshot and replaces the old data with the snapshot data. You can utilize these snapshots to perform time travel queries and roll back data as it existed at a specific point in time or at a specific snapshot.

When you configure a source transformation in a mapping, you can configure the **Time Travel Query** in the advanced source properties. The query fetches the data based on a timestamp or snapshot ID that you specify.

You can use one of the following queries to retrieve the data:

Query by Timestamp

You can query an Iceberg table as it existed at a particular timestamp. You can specify an absolute timestamp or a relative timestamp reference, such as `latest` or `latest-N`.

For example, use the following queries:

For an absolute timestamp, if you want to time travel to July 10, 1986 at 04:20:09, use the following time travel query:

```
TimestampValue=1986-07-10 04:20:09.000 UTC
```

For a relative timestamp, if you want to query based on the second latest recorded timestamp, use the following time travel query:

```
TimestampValue=latest-1
```

Query by Snapshot ID

You can query an Iceberg table by specifying a snapshot ID. Each snapshot has a unique identifier. You can specify an absolute snapshot ID or a relative snapshot version reference, such as `latest` or `latest-N`.

For example, use the following queries:

For an absolute snapshot ID, if you want to time travel to a snapshot with ID 33444444553321, use the following time travel query:

```
SnapshotVersion=33444444553321
```

For a relative snapshot version, if you want to query based on the most recent snapshot, use the following time travel query:

```
SnapshotVersion=latest
```

You can also use the advanced filter option to configure a query for Apache Iceberg or Delta Lake tables. For example, use one of the following queries:

- Query by Timestamp:
  `TimestampValue = '1986-07-10 04:20:09.0 UTC'`
- Query by Snapshot ID:
  `SnapshotVersion = 33444444553321`

Note: When both the **Time Travel Query** advanced property and the advanced filter are configured, the **Time Travel Query** advanced property takes precedence.

Open Table targets in mappings

When you configure a mapping to use an Open Table target, configure the target properties.

Specify the name and description of the Open Table target. Configure the target and advanced target properties for the Open Table object in a Target transformation.

The following table describes the properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection. You can select an existing connection, create a new connection, or define parameter values for the target connection property. If you want to overwrite the target connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Target Type	Type of the Open Table target object. You can choose from the following target types: - **Single Object**. Select to specify a single Open Table object. - **Parameter**. Select to specify a parameter name. You can configure the target object in a mapping task associated with a mapping that uses this Target transformation.
Parameter	A parameter file where you define values that you want to update without having to edit the task. Select an existing parameter for the target object or click **New Parameter** to define a new parameter for the target object. The **Parameter** property appears only if you select parameter as the target type. If you want to overwrite the target object at run time, select the **Allow parameter to be overridden at run time** option. When the task runs, the Secure Agent uses the parameters from the file that you specify in the advanced session properties.

Property	Description
Object	Name of the target object. You can select an existing object from the list or create a target object at run time. For information on creating the target object at run time, see the "Create a target table at run time " on page 30 topic. **Note:** You cannot create a target object at runtime if you configure Hive Metastore catalog in the connection.
Operation	Type of the target operation. Select one of the following operations: - Insert - Update - Upsert - Delete **Note:** You cannot configure the Data Driven operation on an Open Table target.
Update Columns	Select the unique key column as the condition field for the update, upsert, or delete operations in an Open Table target.

The following table describes the advanced properties that you can configure in a Target transformation:

Property	Description
Truncate Target	If enabled, the Secure Agent truncates the target table before loading the data. By default, the check box is not selected.
Enable Transactional Commit	If enabled, the rows are committed periodically during the job based on the commit interval set in the mapping task. By default, this checkbox is not selected, so all the rows are committed at the end of the job. To configure the commit interval, select this checkbox and specify the commit interval in the mapping task. If you do not specify a commit interval, the default value of 10,000 records is used. A lower commit interval might impact the mapping performance. To optimize the performance, commit all the rows at the end of the job or configure a higher commit interval.

Property	Description

Iceberg Advanced Properties	The properties that you want to configure for the Iceberg tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use AWS Glue Catalog with Amazon S3 and the source and target buckets are in different regions, you must specify the Amazon S3 bucket region property in the following format for the update, upsert, or delete target operations:
	`BucketRegion=<Amazon-S3-target-bucket-region-name>`
	In a mapping, when you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<bucket-region-name>`
	In a mapping in advanced mode, when you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket ARN property in the following format:
	`s3.access-points.<bucket-name>=<S3-bucket-ARN>`
	When you use Hive metastore with Amazon S3 and the source and target are in different regions, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<Amazon-S3-target-bucket-region-name>`
	When you use REST Catlog with Amazon S3 and the source and target are in different regions, you must specify the Amazon S3 bucket region property in the following format:
	`restcatalog.iceberg.s3.client.region = <Amazon-S3-source-bucket-region-name>`

Delta Spark Properties*	The properties that you want to configure for the Delta Lake tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<bucket-region-name>`

UpdateMode	Loads data to the target based on the mode you specify. This property applies when you select the **Update** or **Upsert** operation. Select one of the following modes: - Update As Update. Updates records in the target table if the specified unique key column value matches with the incoming column value. - Update Else Insert. Updates records in the target table if the specified unique key column value matches with the incoming column value. If the unique key column value does not match, the mapping inserts a new row with the records.
Pre-SQL	Pre-SQL queries to run before writing data to Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the pre-SQL query fails, the mapping also fails.

Property	Description
Post-SQL	Post-SQL queries to run after writing data to Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the post-SQL query fails, the mapping also fails. If the mapping fails, the post-SQL query is not executed.
Snapshot Retention	Configurations to manage table history and optimize storage by automatically expiring snapshots and deleting data files that are no longer needed. To enable snapshot retention, set the **enable snapshot expiry** property to true for the target table. You can customize the following predefined properties: - Snapshot retention period. Deletes all snapshots created before the specified number of days. The default value is 7d. - Minimum snapshots to retain. Retains the latest number of specified snapshots, even if they're older than the retention period. The default value is 5. - Expiry snapshot id. Deletes a particular snapshot you want to remove. Specify the snapshot ID of the snapshot.
Orphan File Cleanup*	Configurations to remove files that are not referenced by the table metadata. To enable orphan file cleanup, set the **enable orphan file cleanup** property to true for the target table. You can customize the following predefined property: - Orphan file retention period. Deletes all orphan files created before the specified number of days. The default value is 3d.
Rewrite Manifest Files	Configuration to compact data files by combining small manifest files into fewer, larger ones. To enable rewrite manifest files, set the **enable rewrite manifest** property to true for the target table.
*Applies only to mappings in advanced mode.	

Create a target table at run time

You can use an existing target or create a target to hold the results of a mapping. If you choose to create the target, the agent creates the target if it does not exist already when you run the task.

You can create a target at run time only for a AWS Glue catalog or Nessie catalog.
To specify the target properties, perform the following tasks:

1. Select the Target transformation in the mapping.
2. To specify the target, click the **Target** tab.
3. Select the target connection.
4. For the target type, choose **Single Object** or **Parameter**.
5. Specify the target object or parameter.

6. To specify a target object, perform the following tasks:
   a. Click **Select** and choose a target object. You can select an existing target object or create a new target object at runtime.
   b. To create a target object at runtime, select **Create New at Runtime**.
   c. In the **Object Name** field, enter the name of the target table that you want to create. The target table name must be in lowercase.
   d. In the **Catalog Name with Database Name** field, specify the catalog details based on your catalog type:
      - **AWS Glue catalog:** Enter the catalog name and database name in the following format: `<catalogname>/<databasename>`. The database name must be in lowercase.
      - **Nessie catalog:** Enter the branch or reference name and namespace in the following format: `<referencename>/<namespace>`. The namespace must be in lowercase.
   e. In the **Table Location** field, enter the location of the target table data in the following format: `s3://<Bucket_Name>/<Directory_Name>`
   f. Specify the table properties to optimize the table configuration settings.
   g. Click **OK**.

Open Table lookups in mappings

In a mapping, you can configure a Lookup transformation to represent an Open Table object.

You can retrieve data from an Open Table lookup object based on the specified lookup condition.

The following table describes the Open Table lookup properties that you can configure in a Lookup transformation:

Property	Description
Connection	Name of the lookup connection. You can select an existing connection, create a new connection, or define parameter values for the lookup connection property. If you want to overwrite the target connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Source Type	Type of the Open Table lookup object. Select one of the following lookup object types: - **Single Object**. Select to specify a single Open Table object. - **Parameter**. Select to specify a parameter name. You can configure the lookup object in a mapping task associated with a mapping that uses this Lookup transformation.

Property	Description
Parameter	A parameter file where you define values that you want to update without having to edit the task. Select an existing parameter for the lookup object or click **New Parameter** to define a new parameter for the lookup object. The **Parameter** property appears only if you select parameter as the source type. If you want to overwrite the lookup object at run time, select the **Allow parameter to be overridden at run time** option. When the task runs, the Secure Agent uses the parameters from the file that you specify in the advanced session properties.
Lookup Object	Name of the Open Table lookup object.
Multiple Matches	The behavior when the lookup condition returns multiple matches. You can select one of the following options: - Return any row - Return all rows - Report error For more information about configuring lookups, see the *Transformations* guide in the Data Integration documentation.

The following table describes the Open Table lookup advanced properties that you can configure in a Lookup transformation:

Property	Description

Iceberg Advanced Properties	The properties that you want to configure for the Iceberg tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	In a mapping, when you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<bucket-region-name>`
	In a mapping in advanced mode, when you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket ARN property in the following format:
	`s3.access-points.<bucket-name>=<S3-bucket-ARN>`
	For more information about Amazon S3 bucket ARN, see the Create the ARN for your AWS S3 bucket Knowledge Base article.
	When you use Hive metastore with Amazon S3, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<Amazon-S3-bucket-region-name>`
	When you use REST Catlog with Amazon S3, you must specify the Amazon S3 bucket region property in the following format:
	`restcatalog.iceberg.s3.client.region=<Amazon-S3-bucket-region-name>`

Delta Spark Properties*	The properties that you want to configure for the Delta Lake tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use AWS Glue Catalog with Amazon S3 and they are in different regions, you must specify the Amazon S3 bucket region property in the following format:
	`BucketRegion=<bucket-region-name>`

Pre-SQL	Pre-SQL queries to run before reading data to Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the pre-SQL query fails, the mapping also fails.

Property	Description
Post-SQL	Post-SQL queries to run after reading data to Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the post-SQL query fails, the mapping also fails. If the mapping fails, the post-SQL query is not executed.
*Applies only to mappings in advanced mode.	

Dynamic schema handling

You can choose how Data Integration handles changes that you make to the data object schemas. To refresh the schema every time the mapping task runs, you can enable dynamic schema handling in the task.

When you use Apache Iceberg tables as the Open Table format, you can configure dynamic schema handling.

The schema change includes the newly added fields within the data object.

You can configure schema change handling on the **Runtime Options** page when you configure the task.

The following table describes the schema change handling options:

Option	Description
Asynchronous	Default. Data Integration refreshes the schema when you edit the mapping or mapping task, and when Informatica Intelligent Cloud Services is upgraded.
Dynamic	Data Integration refreshes the schema every time the task runs. You can choose from the following options to refresh the schema: - Alter and Apply Changes. Data Integration adds new fields from the source schema to the target schema. - Drop current and recreate. This option only applies to a target created at run time. Drops the existing target table and then recreates the target table at runtime using all the incoming metadata fields from the upstream transformation. You can't use this option for partitioned tables. - Don't apply DDL Changes. Data Integration does not apply the schema changes to the target.

For more information, see the "Schema change handling" topic in *Tasks* in the Data Integration help.

Rules and guidelines for dynamic schema handling

Consider the following rules and guidelines when you enable dynamic schema change handling:

- When you use Apache Iceberg Open Table format and create a target table at runtime in a mapping, ensure that the table name and database name contain only lowercase characters.

- When you enable dynamic schema handling for Apache Iceberg tables, the mapping fails in the following scenarios with the source object:
  - Rename the column.
  - Update the data type.
  - Update the precision.

Rules and guidelines for mappings and mapping tasks

You need to follow certain rules and guidelines to configure tasks that uses Open Table Connector.

General rules and guidelines

- You can't use uncached lookup in the Lookup transformation.
- You can't preview data for a source object when you use a REST or Nessie catalog.

Mappings

- Ensure that the Apache Iceberg tables are in Parquet format.
- You can't use the **Ends With** or **Contains** operators in a simple filter to read from or write to Apache Iceberg tables.
- When you run a mapping with an Open Table connection to read from or write data to Apache Iceberg tables available in AWS Glue Catalog, the mapping fails if AWS Glue Catalog and Amazon S3 bucket are in different regions.
- You can't parameterize a simple filter.
- It is not recommended to modify and rerun an existing mapping that was originally configured for insert operation to perform update, upsert, or delete operations on partitioned tables, as the existing target manifest file might lack the partition metadata. If you need to perform update, upsert, or delete operations with existing mappings, you first need to reingest the data by enabling the Truncate Target property in the mapping. This ensures that the partition metadata is refreshed and stored before performing other target operations.
- Open Table Connector uses Iceberg's Merge-on-Read (MOR) strategy for update, upsert, and delete operations. Frequent use of these operations generates equality delete files and data files that can degrade the reader performance over time on that table. To maintain optimal reader performance, run regular Apache Iceberg table maintenance optimizations. For example, Athena's `OPTIMIZE` and `VACUUM` commands, Snapshot Expiration, Manifest Rewriting, and Orphan File Cleanup.

Mappings in advanced mode

- A mapping fails at run time when you specify a simple filter with **Starts With**, **Ends With**, or **Contains** operators to retrieve data. To avoid this issue, configure an advanced filter with a `like` query to filter query results based on patterns instead of exact textual matches.
- When you configure an advanced data filter for a field of Timestamp data type, the filter ignores the time and considers only the date.
- When you use a simple filter on the date or timestamp data type columns, the task fails at run time with the following error:

```
An error occurred during mapping translation: Invalid expression string for filter
condition
```

As a workaround, use the advance filter to filter date and timestamp data type columns from an Open Table source.

- When you use an Upsert target operation and all the condition fields are not mapped in the field mapping, the mapping fails.
- When you read from or write to Apache Iceberg tables in AWS Glue Catalog, the mapping fails if the source and target belong to different AWS Glue catalogs in separate regions.
- When you configure a mapping with an Apache Iceberg table managed by Hive Metastore, ensure the source and target tables use the same storage type. Mapping fails if one table uses Amazon S3 and the other uses Microsoft Azure Data Lake Storage Gen2.
- When you read data from an Apache Iceberg table managed by the Hive Metastore catalog in a mapping, the preview job fails.
- When you read a large volume of data from an Apache Iceberg table managed by the Hive Metastore catalog in a mapping, the data preview for the source table fails.
- When you configure a mapping, ensure that the source and target tables use the same Open Table format. Mapping fails if one table use the Apache Iceberg format and the other uses Delta Lake format.
- The runtime statistics of the Lookup transformation don't show the count of records processed based on the lookup condition. Instead, it shows the count of rows read from the source table.
- When a mapping fails due to rejected or error rows at the target, the runtime statistics for the source and target don't show the correct information.

CHAPTER 4

Data type reference

Data Integration uses the following data types in Open Table mappings and mapping tasks:

- **Open Table native data types**: Open Table data types appear in the source and target transformations when you choose to edit metadata for the fields.
- **Transformation data types**: Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. Transformation data types appear in all transformations in a mapping.
  When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Open Table and transformation data types

The following table lists the Open Table data types that Data Integration supports and the corresponding transformation data types:

Open Table data type	Transformation data type	Supported Open Table format	Description
Bigint	Bigint	Apache Iceberg,Delta Lake	-9223372036854775808 to 9223372036854775807
Binary	Binary	Apache Iceberg, Delta Lake	Variable length binary data.
Boolean	String	Apache Iceberg, Delta Lake	True or false
Date	Date/Time	Apache Iceberg, Delta Lake	Calendar date without timezone and time. 0001-01-01 to 9999-12-31 Precision 29, scale 9
Decimal (Precision, Scale)	Decimal	Apache Iceberg, Delta Lake	- For mappings, the maximum values for precision and scale are 28. - For mappings in advanced mode, the maximum values for precision and scale are 38.

Open Table data type	Transformation data type	Supported Open Table format	Description
Double	Double	Apache Iceberg, Delta Lake	4.94065645841246544e-324d to 1.79769313486231570e+308d, positive or negative.
Float	Double	Apache Iceberg, Delta Lake	1.40129846432481707e-45 to 3.40282346638528860e+38, positive or negative.
Int	Integer	Apache Iceberg, Delta Lake	2^{31} to 2^{31} -1.
Smallint*	Integer	Delta Lake	-32768 to 32767
String	String	Apache Iceberg, Delta Lake	Variable length character data.
Timestamp	Date/Time	Apache Iceberg, Delta Lake	Timestamp without timezone. Jan 1, 0001 A.D. to Dec 31, 9999 A.D. (Precision to the microsecond)
Tinyint*	Integer	Delta Lake	-128 to 127
*Applies only to mappings in advanced mode.			

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